

SQL-Based Financial Behavior Analysis

Case Study 1: Silent High-Value Accounts

The bank has observed that several customers maintain high-value accounts but do not exhibit expected transaction activity. Some accounts appear underutilized, while others show irregular financial behavior. Additionally, certain relationships between customers and accounts may not be fully visible due to incomplete or indirect data connections.

The institution also suspects that some customers demonstrate similar financial behavior patterns, even though they are not directly linked through obvious relational paths.

Your task is to analyze the dataset to identify all unusual activity. Generate a **single query** based on your assumptions and logic. One mandatory calculated field “Account flag” should describe the nature of all the abnormalities found. You may use as many calculated fields as you want to demonstrate your output.

```
USE FinanceLab;
```

```
WITH customer_summary AS (
```

```
    SELECT
```

```
        c.customer_id, c.customer_name, c.city,
```

```
        COUNT(DISTINCT a.account_id) AS total_accounts,
```

```
        COUNT(t.transaction_id)      AS total_transactions,
```

```
        IFNULL(SUM(t.amount), 0)      AS total_amount
```

```
    FROM Customers c
```

```
    INNER JOIN Accounts a ON c.customer_id = a.customer_id
```

```
    LEFT JOIN Transactions t ON a.account_id = t.account_id
```

```
    GROUP BY c.customer_id, c.customer_name, c.city
```

```
)
```

```
SELECT
```

```
    customer_name, city, total_accounts, total_transactions, total_amount,
```

```

CASE

    WHEN total_transactions = 0 THEN 'Dormant Customer'

    WHEN total_transactions >= (SELECT AVG(total_transactions) FROM
customer_summary)

        AND total_amount >= (SELECT AVG(total_amount) FROM
customer_summary)

        THEN 'Highly Engaged Customer'

    WHEN total_amount >= (SELECT AVG(total_amount) FROM customer_summary)

        THEN 'High Value Low Activity'

    WHEN total_transactions < (SELECT AVG(total_transactions) FROM
customer_summary)

        THEN 'Below Average Activity'

    ELSE 'Normal Customer'

END AS customer_behavior

FROM customer_summary;

```

Result Grid Filter Rows: Export: Wrap Cell Content:						
	customer_name	city	total_accounts	total_transactions	total_amount	customer_behavior
▶	Ali Khan	Lahore	2	5	65000.00	Highly Engaged Customer
	Sara Ahmed	Karachi	1	2	35000.00	Highly Engaged Customer
	Bilal Sheikh	Islamabad	2	1	50000.00	High Value Low Activity
	Ayesha Malik	Lahore	1	1	12000.00	Below Average Activity
	Hassan Raza	Karachi	1	0	0.00	Dormant Customer

Result 34 x

Assumptions & Overview

This query uses the **FinanceLab** database to analyze customer activity by combining **Customers**, **Accounts**, and **Transactions** tables. It summarizes each customer's total accounts, total transactions, and total transaction amount, then classifies customers based on their behavior compared to overall averages.

Assumptions

- Every customer has at least one account.
- Every account belongs to one customer.

- Every transaction belongs to one account.
- Customers with no transactions are still included (shown as 0 amount).
- Customer behavior is judged only on transaction count and total amount.
- Average transaction count and amount are used as benchmark values.

Joins Used

- **INNER JOIN (Customers → Accounts):** excludes customers without accounts.
- **LEFT JOIN (Accounts → Transactions):** includes customers with no transactions.

Behavior Categories

- **Dormant Customer:** no transactions.
- **Highly Engaged Customer:** transactions and amount $\geq$ average.
- **High Value Low Activity:** high amount but fewer transactions.
- **Below Average Activity:** transactions below average.
- **Normal Customer:** average behavior.

SQL Concepts Used

CTE, INNER JOIN, LEFT JOIN, COUNT, SUM, IFNULL, GROUP BY, AVG, subqueries, CASE.

4. Case Study 2: Hidden Customer Behavior Network

The bank is conducting a deeper analysis of customer behavior across cities and segments. Initial observations suggest that customer engagement varies significantly, with some users showing frequent financial activity while others appear inconsistent or partially recorded in transactional data.

There is also a possibility that customers across different categories exhibit similar financial behavior patterns that are not immediately visible through direct comparison.

Generate a **single query** to analyze customer behavior. A mandatory calculated field “Customer Behavior” should highlight the behavior of each customer. You may use any other number of fields in your suggested sql.

USE FinanceLab;

```

WITH customer_summary AS (
    SELECT
        c.customer_id, c.customer_name, c.city,
        COUNT(DISTINCT a.account_id) AS total_accounts,
        COUNT(t.transaction_id) AS total_transactions,
        IFNULL(SUM(t.amount), 0) AS total_amount
    FROM Customers c
    JOIN Accounts a ON c.customer_id = a.customer_id
    LEFT JOIN Transactions t ON a.account_id = t.account_id
    GROUP BY c.customer_id, c.customer_name, c.city
),
final_data AS (
    SELECT cs.*,
        (SELECT AVG(total_amount) FROM customer_summary) AS avg_amount
    FROM customer_summary cs
)
SELECT f.*, r.account_id,
    CASE
        WHEN total_transactions = 0 THEN 'Dormant Customer'
        WHEN total_amount >= avg_amount THEN 'High Value Customer'
        WHEN total_transactions < 3 THEN 'Low Activity Customer'
        ELSE 'Normal Customer'
    END AS customer_behavior
FROM final_data f
RIGHT JOIN Accounts r ON f.customer_id = r.customer_id;

```

Assumptions & Overview

This query from **FinanceLab** analyzes customer banking activity by combining **Customers**, **Accounts**, and **Transactions** data. It first summarizes each customer's financial behavior using a CTE and then classifies them into behavior categories.

Assumptions

- Every account is linked to a valid customer.
- Every transaction belongs to an account.
- Customers without transactions are still included (treated as 0 activity).
- Transaction amount is the main indicator of customer value.
- Average transaction amount is used as a benchmark for classification.
- Customer behavior is determined only from transaction count and amount.

Joins Used

- **INNER JOIN (Customers → Accounts)**: ensures only valid customers with accounts are included in summary.
- **LEFT JOIN (Accounts → Transactions)**: includes accounts even if no transactions exist.
- **RIGHT JOIN (Final step with Accounts)**: ensures all accounts are shown, even if customer summary is missing.

CTEs Used

- **customer_summary**: calculates total accounts, transactions, and total amount per customer.
- **final_data**: computes the average transaction amount for comparison.

Customer Classification Logic

- **Dormant Customer**: no transactions.
- **High Value Customer**: $\text{total amount} \geq \text{average amount}$.
- **Low Activity Customer**: fewer than 3 transactions.
- **Normal Customer**: all remaining customers.

SQL Concepts Used

CTE (WITH), INNER JOIN, LEFT JOIN, RIGHT JOIN, COUNT(DISTINCT), SUM, IFNULL, GROUP BY, AVG, subquery, CASE statement.

customer_id	customer_name	city	total_accounts	total_transactions	total_amount	avg_amount	account_id	customer_behavior
1	Ali Khan	Lahore	2	5	65000.00	24200.000000	101	High Value Customer
1	Ali Khan	Lahore	2	5	65000.00	24200.000000	102	High Value Customer
2	Sara Ahmed	Karachi	1	2	35000.00	24200.000000	103	High Value Customer
3	Bilal Sheikh	Islamabad	2	1	50000.00	24200.000000	104	High Value Customer
3	Bilal Sheikh	Islamabad	2	1	50000.00	24200.000000	105	High Value Customer